MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (BICT/BLIS and BSCEICT)

END OF SEMESTER EXAMINATION

ICT 1103 : COMPUTER AND COMMUNICATION TECHNOLOGY

TIME ALLOWED: 3 HOURS **DATE : APRIL 7, 2017**

INSTRUCTIONS

i. This paper contains **Two** pages. Please check.
ii. There are **Five questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question One

Most organisations utilise a variety of information and usually have more than one information system and below is the list of those information systems.

1. Give **three** examples of each of the following:
   a. Executive Support Systems/Executive Information Systems **(3 Marks)**
   b. Management Information Systems **(3 Marks)**
   c. Knowledge Management Systems **(3 Marks)**
   d. Transaction Processing Systems **(3 Marks)**
   e. Office Automation Systems **(3 Marks)**
2. State **four** purposes of doing analysis and requirements capture in Systems Development Life Cycle. **(4 Marks)**

Question Two

1. List **four** distinct parts of Information Technology systems in an organization. **(4 Marks)**
2. Mention and explain **four** impacts of Information Technology in an organization. **(8 Marks)**
3. What is the difference between Information Systems and Information Technology? **(4 Marks)**
4. Give **one** example how Information systems can be used for? **(1 mark)**
5. Mention **four** primary issues in Computer Ethics. **(4 Marks)**
6. Mention and discuss **four** threats of computer and communication systems. **(8 Marks)**

Question Three

1. Explain the following Data Communication Terminologies:
   i. Channel **(2 Marks)**
   ii. Bandwidth **(2 Marks)**
2. What is a computer network? **(2 Marks)**
3. Mention **two** principle benefits of computer networks. **(2 Marks)**
4. Mention **three** classifications of computer networks. **(3 Marks)**
5. Define the term Network Topology. **(2 Marks)**
6. What is the difference between Logical and Physical topologies? **(2 Marks)**
7. Explain any **three** types of topologies in computer networks. **(3 Marks)**

Question Four

1. What is a database? **(2 Marks)**
2. Mention **two** types of Data. **(2 Marks)**
3. Mention **four** database functions. **(4 Marks)**
4. List down **four** features of the Database approach. **(4 Marks)**
5. Mention **two** properties of a File Processing Systems (file based system). **(2 Marks)**

Question Five

1. What are Numbering systems? **(2 Marks)**
2. What is the difference between Encoding and Decoding? **(2 Marks)**
3. Convert the following:
   i. $A2DE_{16}$ to binary number systems **(2 Marks)**
   ii. 11111111_2 to hexadecimal number. **(2 Marks)**
   iii. $E5_{16}$ to octal number system **(2 Marks)**
   iv. 348_8 to decimal number systems. **(2 Marks)**
   v. 121336_8 to hexadecimal number systems. **(2 Marks)**
4. Explain the following numbering systems:
   i. ASCII **(2 Marks)**
   ii. EBCDI **(2 Marks)**
   iii. UNICODE **(2 Marks)**

END OF QUESTION PAPER